Simon Akhter

437-376-2527 | s23akhte@uwaterloo.ca | simonakhter.com | linkedin.com/in/simonakhter | github.com/shababaa

Education

University of Waterloo — Waterloo, Canada
Bachelor of Applied Science in Management Engineering — *Expected Apr 2029*
Relevant Coursework: *Data Structures and Algorithms, Databases and Software Design, Distributed Systems*

Experience

Software Engineer Intern — Sep 2026 – Present
University Health Network - Princess Margaret Cancer Centre — *Toronto, ON*

- Reduced MySQL query latency by 60% for 6,000+ users by fixing N+1 queries and adding composite indexes
- Accelerated API response times by 40% across 15+ endpoints by optimizing database calls and data models
- Cut failed research data updates by 80% using MySQL transactions, validation, and automated migrations
- Automated imports of 50,000+ daily research records, cutting processing time by 35% using a Python ETL pipeline with batched MySQL writes and data validation to prevent duplicate records

Software Engineer Intern — Feb 2026 – May 2026
Zomp — *Toronto, ON*

- Built a self-hosted CMS for 10+ documentation sites, cutting publishing time by 96% using Docker and Azure
- Reduced CI build times by 45% by caching Docker layers and dependencies in GitHub Actions
- Cut failed deployments by 60% by adding health checks and automated rollbacks to Azure deployments
- Developed a centralized search API for 10+ documentation sites, reducing search latency by 55% by implementing full-text indexing, query caching, and optimized backend endpoints to deliver faster, more relevant results
- Streamlined publishing from 8 manual steps to 1 by automating file validation, permissions, and GitHub commits

Software Developer Intern — May 2025 – Aug 2025
Code Ninjas — *Richmond Hill, ON*

- Reduced student check-in latency by 30% across 2 locations by optimizing Express.js APIs and MongoDB indexes
- Enhanced API throughput by 25% by migrating CPU-intensive tasks to Node.js worker threads, preventing event loop blocking and improving backend responsiveness during concurrent check-in and data processing workflows
- Expedited feature delivery from 2 weeks to 5 days by building reusable React components and Express.js APIs

Projects

Market Intelligence | *Go, B+ Trees, Transactions, REST APIs, Docker, GitHub Actions* — Jun 2026

- Built a Go B+ tree database with crash recovery, cutting insertion time by 77% and memory use by 85%
- Lowered stock-price query latency by 86% using composite B+ tree indexes and compressed storage
- Built a Go market API handling 262 requests/s at 93 ms p95 using indexed queries and durable background jobs
- Implemented a restart-safe Go pipeline ingesting 5 years of stock data across 3 assets, using durable checkpoints and optimized B+ tree writes to cut 100-record batch ingestion time by 69%

VectorForge | *Go, Python, React, TypeScript, FastAPI, PyTorch* — Oct 2026

- Designed a Go semantic search engine indexing 7,478 code and document chunks using MiniLM embeddings
- Accelerated vector search by 39× across 523K embeddings using IVF indexing while retaining 94.9% Recall@10
- Boosted search ranking by 8.4% by fine-tuning MiniLM with contrastive learning and hard negatives
- Constructed a persistent IVF index for 100,000 vectors in a Go B+ tree database, reloading centroids and posting lists 53x faster than retraining while preserving identical search results across 200 test queries

Skills

Languages: Go, Python, TypeScript, JavaScript, C++, Java, C#, SQL
Frameworks: React, Next.js, Node.js, Express, FastAPI, Flask, ASP.NET
Developer Tools: MySQL, PostgreSQL, MongoDB, SQLite, Docker, Git, GitHub Actions, AWS, Azure, Linux